Department of Computer Science and Engineering

UE24CS341A Software Engineering

Project Proposal and Synopsis

Online Exam Portal

An online examination system with multiple-choice questions and automatic evaluation.

Version 1.1 - revised for team review
3 October 2026

We propose a browser-based Online Exam Portal for secure, timed, single-answer multiple-choice examinations. Release 1 covers exam preparation, student attempts, automatic evaluation and basic result viewing. This proposal establishes the scope and division of complete functional work for an Agile implementation.

Project and team details

Team number: 06

Name	SRN
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1. Problem statement and objectives

The Online Exam Portal is a web-based system for conducting multiple-choice examinations digitally. It allows instructors to create, manage, and publish MCQ exams, students to attempt timed exams and submit responses, and the system to automatically evaluate answers and generate results. The project focuses on providing a secure, structured, and easy-to-manage examination workflow for both instructors and students.

1.1 Proposed project description

The portal is a new, self-contained academic application using MongoDB, Express, React, and Node.js. Instructors prepare draft exams with single-answer MCQs and publish validated content. Students see eligible published exams, start or resume one timed attempt, save answers, navigate questions and submit. The server enforces the deadline even when the browser closes, evaluates saved responses and persists results. Admins provision accounts and manage roles and account status. Typical users are students with basic browser skills, instructors who prepare assessments, and an administrator who controls account access. A real external customer has not been confirmed.

1.2 Objectives

• Replace the paper-based MCQ workflow with a consistent digital preparation and examination process.

• Enforce role and ownership checks on the server to protect exams, answers, and results.

• Preserve acknowledged answers and deadlines across navigation and page reloads.

• Calculate and store reproducible scores using the published answer key.

• Demonstrate Agile delivery, individual end-to-end contribution, reviewed PRs, CI/CD and security validation.

1.3 Risks

Main risks are deadline and save/submit races, unauthorized access to exams or results, and scope expansion. Mitigate them with server-authoritative deadlines, atomic persistence, tests that reject unauthorized access, and regular scope review.

2. Release 1 scope and functional features

• Users authenticate through login and logout; the system enforces Student, Instructor and Admin permissions.

• Admins create accounts with unique login identifiers and assign or update a single role and active status.

• Instructors create exams with a title, positive duration and optional availability start time; edit or delete owned drafts; view owned exams.

• Instructors add, edit and delete exam-specific MCQ questions and options, choose exactly one correct option, and publish valid exams.

• Students browse published exams, start an eligible exam or resume the same active attempt, and see a remaining-time display.

• Students select and change answers, save them online, navigate previous and next questions and restore saved selections on reload.

• Students submit manually; the server also finalizes expired attempts automatically, including when a browser is closed.

• The system automatically evaluates persisted answers, calculates the score and stores a durable result.

• Students view their own results; instructors view student results only for their owned exams.

2.1 Operating rules

Each student has one attempt per exam; resume does not create a retake or reset the deadline. The availability time controls when an attempt may begin; the duration runs from its server-recorded start. Published exams and answer keys are immutable. Each publishable question has at least two nonblank options and exactly one correct option. Marking is 1 point for correct and 0 for wrong or unanswered. Admin privileges do not automatically grant student or instructor workflows.

2.2 Outside Release 1

Negative marking; AI question generation or proctoring; reusable question banks; course or enrollment management; advanced analytics; CSV export; offline synchronization; mobile applications. Public self-registration, retakes and answer-key disclosure are not required. Account provisioning is the minimum administration needed for the stated three-role scope.

3. Plan of work and product ownership

The following functional allocation reflects the three-member team plan and is separate from the documentation split. All three members contribute to UI, API, persistence, tests and reviews within their functional work. The proposed schedule must be reconciled with the instructor's official deadlines.

Member	Complete functional ownership	Quality responsibility
Tanusha Prakash	**Exam and Question Management:** Create, edit and delete owned draft exams; configure title, description, start time and duration; manage MCQs and options; select the correct option; publish valid exams.	Validate exam content and ownership; prevent unauthorized changes; preserve published exams and answer keys.
Guntupalli Manaswi	**Exam Attempt:** View published exams; start or resume an eligible attempt; answer and navigate questions; display remaining time; save responses; support manual submission and automatic submission on timeout.	Preserve saved answers; enforce server deadlines; prevent duplicate attempts, late answer changes and duplicate submissions.
Vidimalla Guru Teja	**Evaluation and Results:** Evaluate submitted responses; calculate and store scores; provide student result viewing and instructor result viewing for owned exams.	Ensure repeatable scoring; prevent duplicate results; restrict result access to authorized users.

Proposed short-term plan

Window (2026)	Planned outcomes
3 October	Review and finalize the project proposal and SRS; Tanusha continues planning and design drafts.
4 October	Guntupalli Manaswi completes the initial field-layout workbook and RTM for review; unavailable implementation and test references remain Pending.
5 to 11 October	Sprint 1: review requirements; configure MongoDB, Express, React and Node.js (MERN); implement login, role checks and account administration.
12 to 18 October	Sprint 2: implement owned draft exams, MCQs and publication; validate the instructor workflow.
19 to 25 October	Sprint 3: implement timed attempts, answer persistence, navigation, manual submission and automatic submission on timeout; verify concurrent and deadline cases.
26 October to 1 November	Sprint 4: implement evaluation and result viewing; integrate, test, review, deploy and update traceability.

Shared responsibilities: All three members contribute to authentication and role-based access, account administration, MongoDB database integration, security, testing, code reviews, CI/CD, documentation and requirement traceability. Individual contributions are tracked through Jira tasks and Git commits.

3.1 Delivery process and tools

Use an Agile backlog in Jira, small story-based branches, PR reviews, sprint demonstrations and retrospectives. GitHub holds source and documents. GitHub Actions provides CI, Docker for containerization, Jest and Supertest for API/unit tests, React Testing Library for UI tests, and SonarQube or an equivalent static analyzer. Versions, deployment target and measurable test thresholds will be confirmed in the project plan and design.

Done requires scope-consistent functionality, server validation, relevant tests, teammate review and updated documentation. Each member implements and explains a complete function and retains attributable Git commits. Testing includes unit, integration, system and security checks; actual results and coverage are recorded after execution.